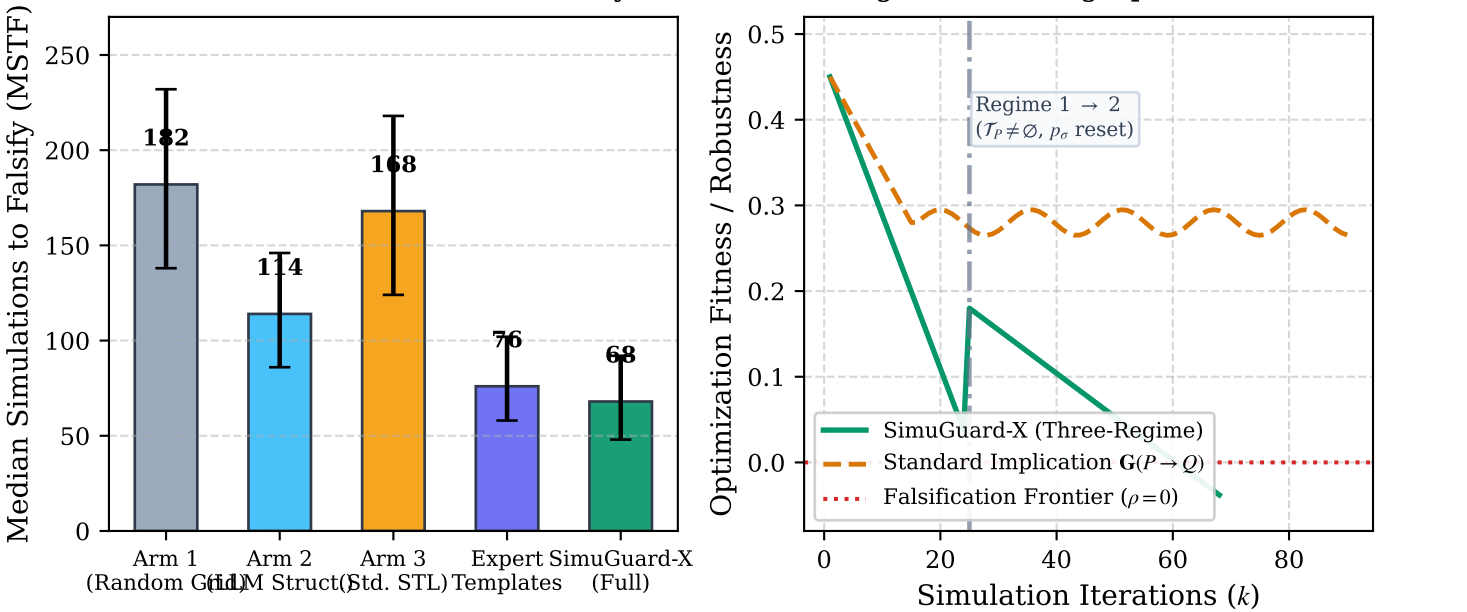
(a) Factorial Ablation: Search Efficiency (MSTF)
Median Simulations to Falsify (MSTF)
182
114
168
76
68
Arm 1 (Random Grid)
Arm 2 (LLM Struct.)
Arm 3 (Std. STL)
Expert Templates
SimuGuard-X (Full)
(b) Convergence: Lexicographic vs. Standard STL
Optimization Fitness / Robustness
Regime 1 → 2
($\mathcal{T}_P \neq \emptyset$, p_σ reset)
SimuGuard-X (Three-Regime)
Standard Implication $\mathbf{G}(P \rightarrow Q)$
Falsification Frontier ($\rho = 0$)
Simulation Iterations (k)